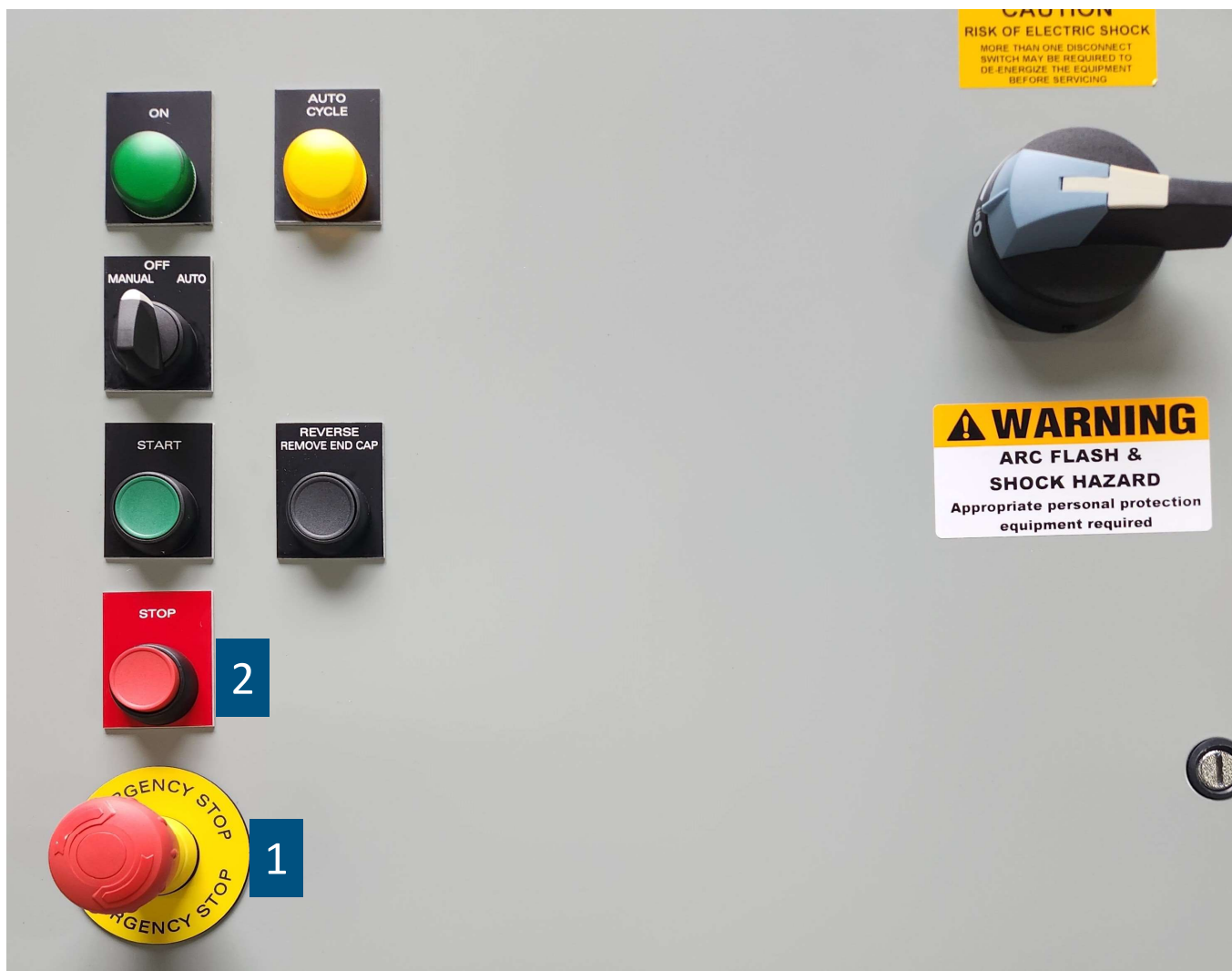


10.0: Electrical Panel Overview

NOTICE

The start up of control panels that contain customer specified options, PLC units, or HMI units, come with a custom, job specific manual from Hapman. That manual should be referred to when performing startup.

1. **Emergency Stop Button (E-stop)**-Pressing this button should instantly stop the unit from running and prevent it from being restarted while depressed.
2. **Stop Button** -This button will stop the unit but will not prevent the restart of the unit like the E-stop.

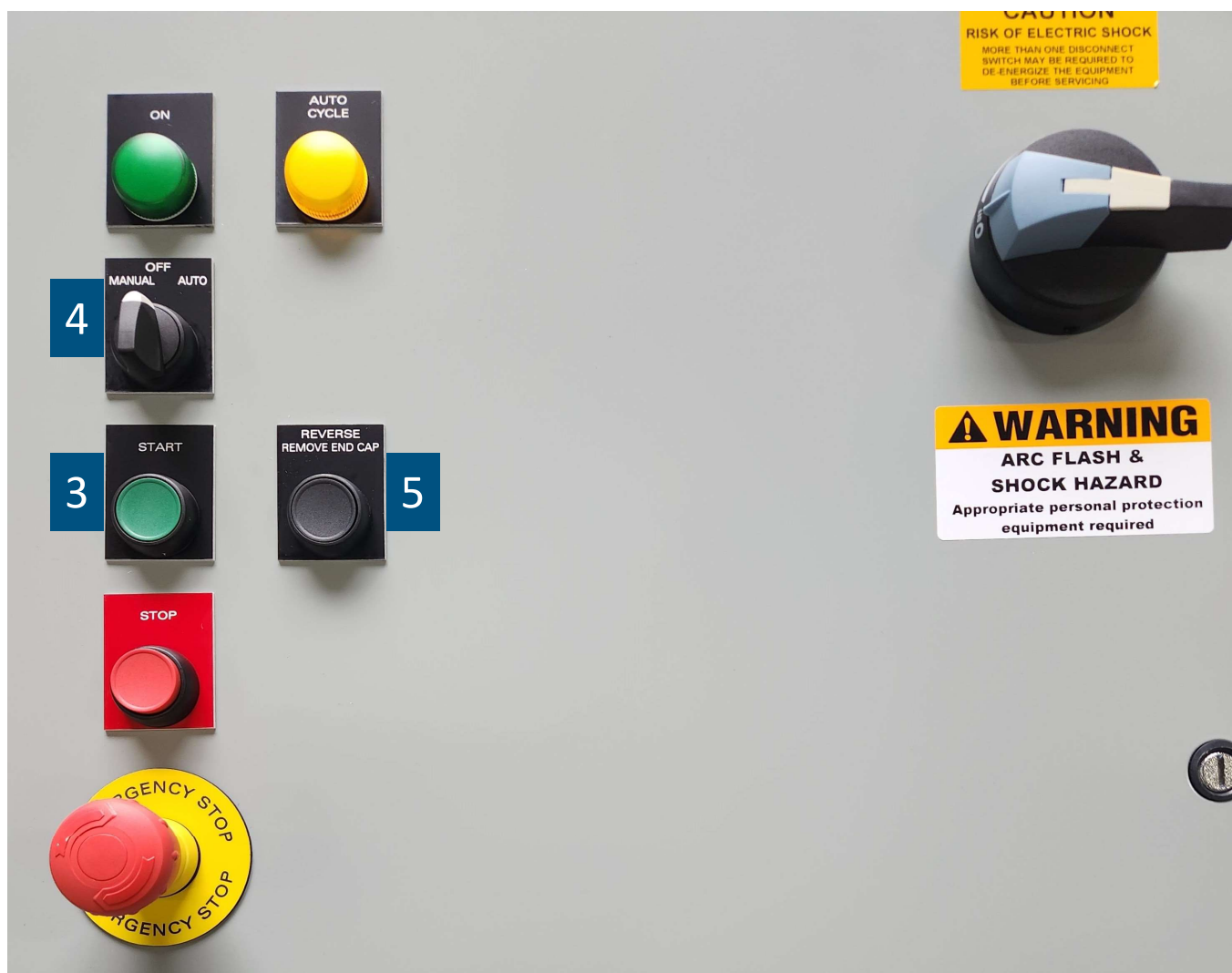


10.0: Electrical Panel Overview

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| 3. | Start Button – This button starts the unit in both Auto and Manual modes. |
| 4. | Run Mode Selection Switch – This selector switch lets the operator choose between operating the unit in Auto Mode or Manual Mode. Off mode will prevent the unit from starting. |
| 5. | Reverse Button – This is a momentary push button (meaning it must be held down to run and once released will stop) |

NOTICE

*Before using the reverse button make sure the hopper boot end cap is removed and the unit is in manual mode. This feature should **ONLY** be used to clean out residual material from the hopper.*



10.0: Electrical Panel Overview

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| 6. | Rotary Disconnect – This handle turns On/OFF the high voltage supply power to the unit. The white tab can be pulled out and used for Lock Out Tag Out (<i>indicated by the red arrow below</i>). |
| 7. | Auto Cycle Light (yellow) – This light comes on when the unit is in Auto mode and the start button has been pressed. |
| 8. | ON Light – This light comes on in Auto or Manual modes, once the VFD (or motor starter on some units) is pulled in and running. |



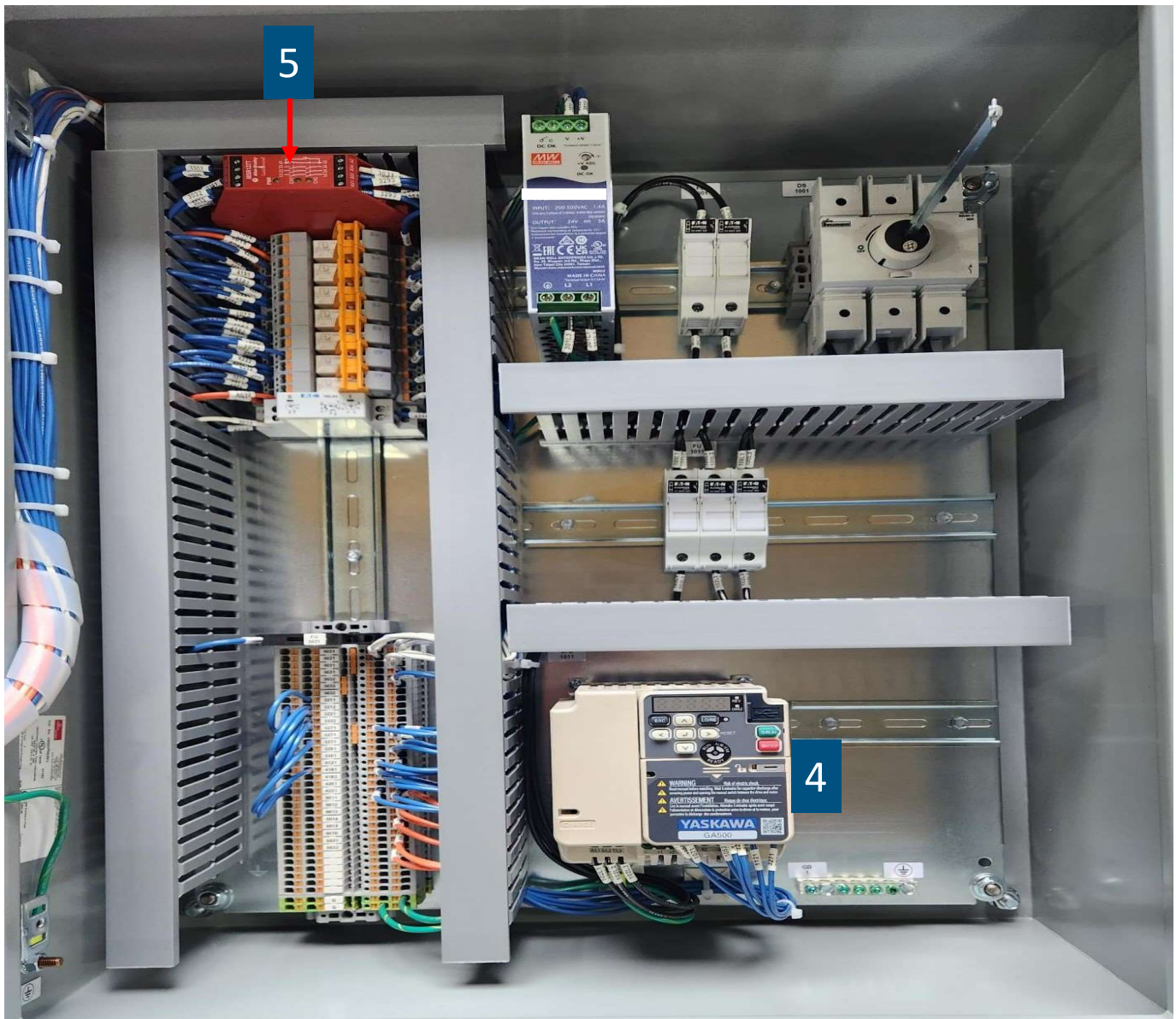
10.0: Electrical Panel Overview

1. **Rotary Disconnect** – This turns On/OFF the high voltage supply power to the unit.
2. **Fuses**- In line circuit protection
3. **24 VDC Power Supply** – Converts incoming voltage to 24 vdc control voltage.



10.0 Electrical Panel Overview

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| 4. | VFD – The frequency drive controls the speed of the motor and allows speed adjustments. <i>(Some control panels will have fixed speed motor contactors in place of a VFD, based on options selected during the sales process)</i> |
| 5. | Safety Control Relay -This control relay is where all safety device wire back to. This ensures that when an E-stop or safety sensor is disengaged, control power is dropped out. |



10.0: Electrical Panel Overview

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| 6. | Control Relays - These relays control the equipment function throughout the system by being turned on or off. |
| 7. | Terminal Strip - This is where all incoming external wiring and existing internal control wiring is landed/terminated. |
| 8. | On Delay Timer - Used to delay the start of the conveyor when using an exterior signal to tell the Helix to turn on or off. |

